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Case Study - Top Generative AI Tools

Generative AI tools harness AI models to produce unique content across mediums such as text, audio, and visual formats. They rely on natural language processing, neural networks, and deep learning algorithms. Their effectiveness is linked to the quality and volume of training data they are fed. Let's explore some of the top GEN AI Tools.

1. GPT-4 by OpenAI :

Attributes : Enhanced creativity and accuracy; Large multimodal model; Image and text input but only text outputs.

Strengths : Improved stability; High test scores; Robust safety measures.

Drawbacks : No public access to image inputs; Concerns about safety and stability; Persistent bias issues.

2. ChatGPT by OpenAI :

Attributes : Retains conversation history; Designed for human dialogue; API for developers; New AI plugins for internet browsing and coding.

Strengths : Accessible and free for general users; API lets developers integrate into their platforms.

Drawbacks : Limited post-2021 data access; Occasional technical glitches; Can produce erroneous or offensive content.

3. AlphaCode by DeepMind :

Attributes: Transformer-based model with a strong focus on Python and C++; Utilizes large-scale code repositories for training.

Strengths : Highly competitive against human coders; Freely accessible on GitHub.



Drawbacks : Outperforms only about half of human competitors; Demands substantial computing resources; Around 4% false positive rate.

4. GitHub Copilot :

Attributes : Transforms language prompts into code suggestions; Offers organization-wide policy management.

Strengths : Grounded on rich data from OpenAI Codex and GitHub; Free trial available.

Drawbacks : Limited support for new APIs or coding libraries; Potential data privacy issues.

5. Bard by Google :

Attributes : Built on LaMDA transformer model; Features programming and software development capabilities.

Strengths : Ethically developed; Constantly updated with web information.

Drawbacks : No conversational history feature; Limited to English language.

6. Cohere Generate :

Attributes : Focuses on marketing and sales content; Adaptable to various cloud environments.

Strengths : Efficient at basic content generation tasks; Easy-to-use API.

Drawbacks : Limited applicability outside sales and marketing; Expensive embedding for some models; Less accurate than some competitors.

7. Claude by Anthropic :

Attributes : Emphasizes safety and conversational tone; Offers Slack integration and supports many languages.

Strengths : Comprehensive context understanding; Uses "red team" prompts for safety.



Drawbacks : Absence of free version; Cannot access the web.

8. Synthesia :

Attributes : AI-driven video creation from scripted prompts; Multiple integrations available.

Strengths : Supports 120+ languages; User-friendly interface.

Drawbacks : Occasional authoring glitches; Potential audio quality issues in basic plans.

9. DALL-E 2 by OpenAI :

Attributes : Uses language inputs for image outputs; Offers features like outpainting and inpainting.

Strengths : Higher resolution than DALL-E; Improved safety protocols.

Drawbacks : Users now need to purchase credits; Can still produce inappropriate content.

10. Code Whisperer by AWS :

Attributes: Quick code generation via language prompts; Free and Paid tiers.

Strengths: Boosts productivity with instant suggestions; aligns with best practices; multi-language & IDE support; AWS-optimized.

Drawbacks: Not a complete human coding substitute; challenges with complex tasks; accuracy concerns; potential open-source legal issues; security risks.

To conclude, these AI tools hold transformative potential across industries. Nonetheless, they remain works-in-progress with occasional errors. Relying solely on them without human judgment is not advisable.

References:

<https://www.eweek.com/artificial-intelligence/generative-ai-apps-tools/>